



UNIVERSIDAD DE GRANADA

Faculty of Science

MCNP Simulation of a PWR Reactor Assembly: Parametric Study of Moderation and Enrichment

Author:

Marco Fumagalli

Subject:

Tecnología Nuclear

Academic Year 2025/2026

June 5, 2026

Contents

Mini PWR-like Assembly Simulation	2
1 Theoretical Background	2
1.1 Chain reaction and the four-factor formula	2
1.2 Lattice pitch and the moderation curve	2
1.3 Fuel enrichment and reactivity saturation	3
1.4 Aim of this work and reference benchmark	3
2 Methods and Setup	4
2.1 Geometry	4
2.2 Materials	5
2.3 Boundary conditions and physics options	5
2.4 Simulation settings	5
3 Validation of the baseline deck	6
4 The 3×3 assembly at the reference PWR pitch	7
5 Effect of the lattice pitch	7
5.1 The moderation curve	7
5.2 Estimation of the optimal pitch	8
5.3 Comparison with the commercial PWR design	8
6 Effect of moderator density (simplified temperature approach)	9
7 Neutron Spectrum and Moderation Regimes	9
8 Effect of the fuel enrichment	11
8.1 Setup of the enrichment scan	11
9 Master Summary	12
10 Conclusions	12
A MCNP6 input decks used in this work	13
A.1 Reference deck (assembly_p126.txt)	14
A.2 Pitch parametric study (Section 5)	14
A.3 Enrichment parametric study (Section 8)	14
A.4 Spectrum tally addition (Section 7)	15
A.5 Moderator temperature variation (Section 6)	15

Mini PWR-like Assembly Simulation

1 Theoretical Background

1.1 Chain reaction and the four-factor formula

A nuclear reactor relies on a self-sustaining chain reaction: each fission of a heavy nucleus releases two or three fast neutrons, of which a sufficient fraction must, on average, induce a new fission in order to keep the reaction going. Because the fission cross-section of ^{235}U is more than two orders of magnitude larger at thermal energies (~ 0.025 eV) than at the few-MeV energies at which neutrons are born, the dominant family of commercial reactors moderates its neutrons inside a hydrogen-rich medium, typically light water, before letting them encounter the next fuel pin. The whole physics of a thermal reactor sits in this competition between moderation, parasitic absorption and geometric leakage, and is conveniently summarised by the four-factor formula [1, 2]

$$k_{\infty} = \eta_F \varepsilon p f, \quad (1)$$

where η_F is the reproduction factor (the average number of fission neutrons produced per thermal neutron absorbed in the fuel), ε the fast-fission factor (the small bonus given by fast fissions in ^{238}U), p the resonance escape probability (the probability that a neutron is moderated past the ^{238}U resonances without being captured), and f the thermal utilisation factor (the fraction of thermal neutrons absorbed in the fuel rather than in the moderator or in structural materials). In a finite system the leakage out of the boundary further reduces the multiplication, so that the effective multiplication factor is

$$k_{\text{eff}} = k_{\infty} p_L, \quad (2)$$

with $p_L < 1$ the non-leakage probability.

1.2 Lattice pitch and the moderation curve

The geometric arrangement of the fuel pins, and in particular the *lattice pitch* p , defined as the centre-to-centre distance between adjacent pins, has a strong, non-monotonic effect on k_{∞} via the four factors of equation (1). At small pitch the volume of moderator surrounding each pin is too small, the spectrum stays hard¹, the resonance escape probability p collapses because of insufficient slowing-down between fuel encounters, and the chain reaction loses neutrons to ^{238}U capture before any of them become thermal. As the pitch is increased the spectrum thermalises rapidly and k_{∞} rises steeply. Beyond a certain optimum, however, the additional water starts contributing more to parasitic absorption² (mostly via radiative capture in ^1H) than to useful moderation; the thermal utilisation factor f then begins to dominate the loss budget and k_{∞} falls again on the over-moderated side. The net result is the well-known “moderation curve”, a concave function of p with a single maximum.

¹The neutron energy spectrum, i.e. the distribution of neutron flux as a function of energy, is conventionally called *hard* when fast-energy neutrons ($E \gtrsim 0.1$ MeV) dominate the population, and *soft* (or thermalised) when the thermal component ($E \lesssim 0.5$ eV) dominates. The three regimes are explicitly shown in Figure 3 of Section 7.

²By *parasitic absorption* we mean any neutron capture event that does not produce a new fission and therefore subtracts neutrons from the chain reaction. The main parasitic channels in a clean $\text{UO}_2/\text{H}_2\text{O}$ lattice are radiative capture in ^{238}U at epithermal resonance energies and (n, γ) capture in ^1H at thermal energies.

1.3 Fuel enrichment and reactivity saturation

A second physical lever which is just as important as the geometry is the *fuel enrichment*, defined as the fraction of ^{235}U in the uranium component of the fuel. Increasing the enrichment increases the reproduction factor η_F , because a larger fraction of the neutrons absorbed in the fuel ends up producing a fission rather than a sterile capture in ^{238}U . The dependence of k_∞ on the enrichment is, however, not linear and saturates at high enrichments, since the contributions of the other factors (ε , p , f) approach their asymptotic values once the ^{238}U content is significantly reduced.

1.4 Aim of this work and reference benchmark

In this work we present a Monte Carlo characterisation of a small, idealised PWR-like fuel assembly that reproduces both dependencies described above and benchmarks the resulting lattice physics against the Westinghouse 17×17 commercial design. The starting point is the example deck `Fision.txt` provided as part of the course material [6], which describes a fixed-source problem with two cylindrical rods of *metallic uranium* immersed in light water. This deck is first run unchanged in order to validate the workflow, and is then extended to a regular 3×3 lattice of UO_2 fuel pins on which a proper criticality calculation (in MCNP’s `kcode` mode) is performed. By imposing reflective boundary conditions on the unit cell, the geometric leakage is suppressed ($p_L = 1$ in the four-factor formula of equation (1)) and the multiplication factor returned by MCNP coincides directly with the infinite-medium multiplication factor k_∞ . Its dependence on the pitch is studied parametrically over a range that spans the under-moderated, the optimal and the over-moderated regimes. A complementary parametric study of the fuel enrichment, performed at the reference pitch $p = 1.26$ cm, completes the characterisation of the lattice.

As mentioned above, throughout the analysis we adopt the design parameters of a Westinghouse 17×17 PWR fuel assembly³ as a real-world benchmark. The key design parameters of the Westinghouse 17×17 that are explicitly reproduced in our 3×3 unit cell are summarised below:

Table 1: Key design parameters of the Westinghouse 17×17 PWR fuel assembly that are explicitly reproduced in our 3×3 unit cell.

Parameter	Symbol	Value	Note
Lattice pitch	p	1.26 cm	centre-to-centre distance between adjacent fuel rods
Fuel-pellet radius	r_p	0.41 cm	UO_2 sintered ceramic pellet
Fuel-pellet density	ρ_{UO_2}	10.40 g/cm ³	typical of sintered ceramic pellets
^{235}U enrichment	–	3%	reference run; varied parametrically in Sec. 8
Moderator density	$\rho_{\text{H}_2\text{O}}$	0.71 g/cm ³	PWR-hot conditions ($T \simeq 600$ K, 15 MPa); see Sec. 6

With this choice of parameters the calculated value of k_∞ and the location of the moderation optimum can be directly compared with the corresponding behaviour of a working commercial PWR, providing a quantitative test of how close our idealised lattice model comes to industrial practice.

³The Westinghouse 17×17 is by far the most widespread fuel-assembly design for modern commercial PWRs worldwide.

2 Methods and Setup

The simulations are run with MCNP6.2 [3] on the dedicated **neutron-Lenovo-S500** server provided for the course, using the ENDF/B-VII.1 cross-section library at room temperature [4]. The geometry, the cross-section libraries and the source cards of the original **Fision.txt** example are inherited as the starting point of the work; the geometric extension to a 3×3 assembly, the conversion of the fuel material from metallic uranium to UO_2 ceramic, and the switch from a fixed-source to a **kcode** criticality (multiplication-factor) calculation are the structural changes introduced. This conversion makes the simulation directly representative of an actual LWR fuel pin: UO_2 ceramic is the universal fuel form of commercial PWR and BWR cores, thanks to its high melting point ($\sim 2865^\circ\text{C}$) and chemical inertness with the moderator. The lower fuel density (10.40 vs 19.0 g/cm^3) and the addition of oxygen reduce the fissile uranium per unit volume, but oxygen is essentially transparent to neutrons in the energy range of interest (its (n, γ) capture cross-section is below 10^{-4} barn at thermal energies), so the change is purely an upgrade of the physical realism of the model.

2.1 Geometry

A schematic of the cross-section of the simulated assembly at the reference pitch $p = 1.26\text{ cm}$ is shown in Figure 1.

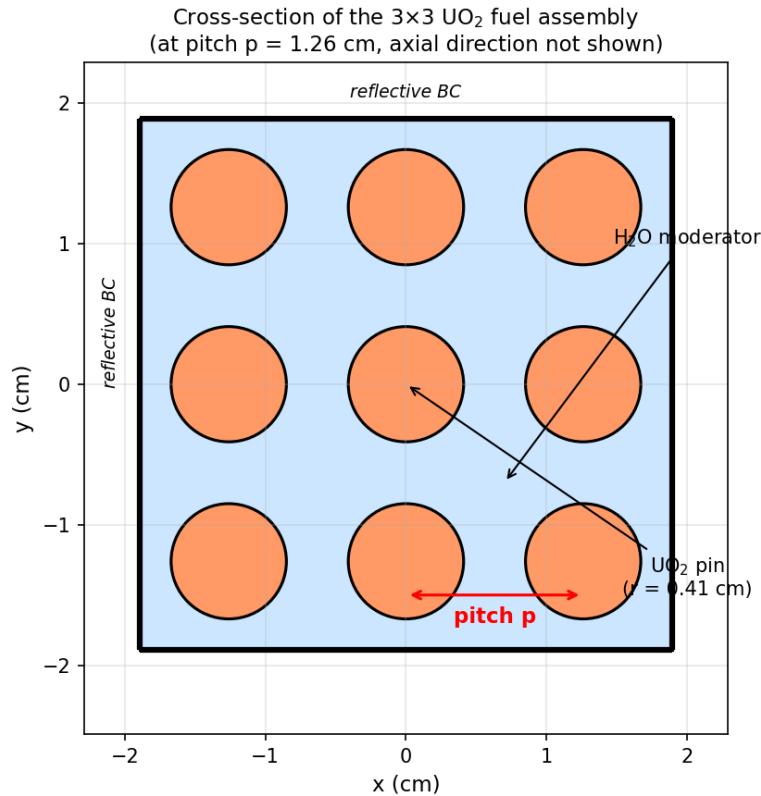


Figure 1: Cross-section of the simulated 3×3 UO_2 fuel assembly at the reference pitch $p = 1.26\text{ cm}$ (axial direction not shown). The thick black lines mark the reflective boundary conditions imposed on the four lateral faces of the cubic unit cell; identical reflective conditions are imposed on the two axial faces, so that the calculation effectively simulates an infinite repeating lattice and returns the infinite-medium multiplication factor k_∞ .

The fuel is modelled as a regular array of nine cylindrical UO_2 pins of radius $r_p = 0.41$ cm and active height $h = 10$ cm ($-5 \leq z \leq +5$ cm), with the pin axes parallel to the z -direction and centred on the nine nodes of a square 3×3 grid. The intra-assembly volume not occupied by fuel is filled with light water at standard density. The unit cell is enclosed by a cubic box whose half-side is deliberately set equal to $1.5p$. With this choice each edge pin sits at exactly $p/2$ from the nearest face of the box. The six reflective boundary conditions then act as mirrors that generate phantom pins at distance p from the real edge pins, matching the lattice spacing exactly. The unit cell therefore tiles periodically in all three directions, and the neutrons effectively see an infinite three-dimensional lattice. The geometric non-leakage probability is exactly $p_L = 1$ and the multiplication factor returned by MCNP coincides directly with the infinite-medium multiplication factor k_∞ .

2.2 Materials

The fuel material is uranium dioxide enriched in ^{235}U with a density of 10.40 g/cm^3 typical of sintered ceramic pellets; the moderator is light water at standard density. The reference fuel composition adopted in Sections 3–5 corresponds to a standard 3% enrichment in ^{235}U , with mass fractions as listed in Appendix A; in Section 8 the enrichment is varied between 1.5% and 20%, with the corresponding mass fractions recomputed accordingly. Table 2 summarises the materials used.

Table 2: Materials adopted in the simulations. Mass fractions are referred to the corresponding compound; the U-235/U-238 split is shown for the reference 3% enrichment used as the baseline configuration.

Material	Isotopes	$\rho \text{ (g/cm}^3\text{)}$	Mass fractions (3% enr.)
UO_2 (3% enr.)	^{235}U , ^{238}U , ^{16}O	10.40	0.0265 / 0.8549 / 0.1187
H_2O	^1H , ^{16}O	1.00	0.1119 / 0.8881

The isotopic composition is reported explicitly because the neutronic behaviour of different isotopes of the same element can be radically different: for instance, ^{235}U undergoes fission upon absorbing a thermal neutron and is the actual fissile component of the fuel, while ^{238}U is primarily a neutron absorber in the same energy range, and the moderating power of light hydrogen (^1H) is completely different from that of its heavier isotope deuterium (^2H , used in heavy-water reactors).

2.3 Boundary conditions and physics options

The transport calculation is run in pure neutron mode (`mode n`) with analog physics (`phys:n`). Reflective boundary conditions are imposed on all six faces of the cubic unit cell by prefixing the corresponding surfaces with an asterisk in the MCNP input deck; this choice, together with the geometric prescription described above on the size of the box, makes the simulation equivalent to that of an infinite repeating lattice and is the standard methodological approach for lattice-physics studies in reactor design [5].

2.4 Simulation settings

The model adopted here is intentionally idealised, in the sense that it omits the neutron-absorbing elements that would normally be present in a real PWR core (either

added by design to control reactivity, or accumulated as by-products of operation). The value of k_∞ computed by the simulations therefore represents the upper bound of the reactivity achievable by this lattice configuration.

The criticality calculation in MCNP uses the card `kcode 10000 1.0 30 300`, whose four parameters have the following meaning:

- 10000 – number of neutron histories simulated in each cycle of the iterative procedure;
- 1.0 – initial guess for the multiplication factor (we start by assuming the system is exactly critical, and let MCNP correct this estimate as the iterations proceed);
- 30 – number of *inactive* cycles, discarded from the average: in the first iterations the spatial distribution of fission sources is still relaxing towards its self-consistent shape, so the corresponding k estimates are biased;
- 300 – number of *active* cycles, used to compute the final mean value of k_∞ and its statistical uncertainty.

A total of 3.3×10^6 neutron histories is therefore simulated in every run. The initial neutron source is specified through the `ksrc` card, with one source point placed at the centre of each fuel pin (updated consistently with the lattice pitch). This initialisation gives the iterative procedure a head start – the source distribution is already close to its self-consistent shape from the first cycle – and justifies the conservative choice of 30 inactive cycles.

3 Validation of the baseline deck

Before performing any new calculation, the original `Fision.txt` deck was run unchanged on the host machine. The purpose of this preliminary step is methodological rather than physical: we want to verify, before introducing any modification of our own, that the MCNP installation, the cross-section libraries and the file paths on the server are correctly set up, and that the example deck provided with the course material produces a statistically converged result as it should.

The deck computes the fission reaction rate in one of the two uranium rods, using an `F4` flux tally combined with an `FM` multiplier card that converts the average flux into a reaction rate per source neutron. The simulation returns:

$$R_f = (6.40 \pm 0.25) \times 10^{-3} \quad \text{fissions per source neutron.} \quad (3)$$

The numerical value of R_f is not directly relevant for the rest of this work, since the geometry of `Fision.txt` is very different from the 3×3 assembly studied in the following sections. What matters is that the simulation runs without error, returns a physically sensible value (in the small geometry of the example most source neutrons either escape or are absorbed without fissioning, consistent with the low value of R_f) and passes all ten statistical convergence checks of the tally with a relative error of 4%, well within the conventional 10% target. The baseline workflow is therefore considered validated, and we proceed to its extension.

4 The 3×3 assembly at the reference PWR pitch

The minimal extension of the `Fision.txt` geometry that allows a meaningful study of the lattice physics is the periodic 3×3 arrangement described in Section 2, equipped with the six reflective boundary conditions that turn it into the unit cell of an infinite repeating lattice. As reference configuration we adopt the lattice pitch of the Westinghouse 17×17 fuel assembly introduced in Section 1, $p = 1.26$ cm, while keeping all the other geometric and material parameters as described in Section 2. The multiplication factor obtained from the corresponding `kcode` run, after approximately 3×10^6 active neutron histories distributed over 300 cycles, is

$$k_{\infty} = 1.41856 \pm 0.00031. \quad (4)$$

The relative error on k_{∞} is 0.022%, more than three orders of magnitude smaller than the value of k_{∞} itself. This result is in good qualitative agreement with the typical $k_{\infty} \simeq 1.30$ – 1.45 values quoted in the literature for 3% enriched UO_2 in light water at room temperature [2].

5 Effect of the lattice pitch

5.1 The moderation curve

In the second part of the work the lattice pitch is varied parametrically over five values, $p \in \{0.90, 1.10, 1.26, 1.50, 2.00\}$ cm, while every other physical and numerical setting is left unchanged. The lower limit $p = 0.90$ cm is dictated by the geometric constraint $p > 2r_p = 0.82$ cm: shorter pitches would make adjacent fuel pins overlap, which is unphysical. The five k_{∞} values returned by the corresponding `kcode` runs are reported in Table 3 and plotted in Figure 2.

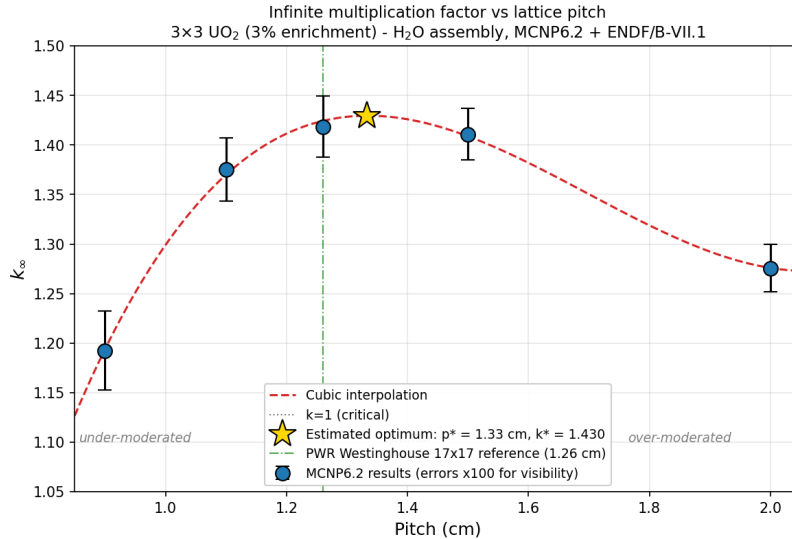


Figure 2: Infinite-medium multiplication factor k_{∞} as a function of the lattice pitch. Statistical error bars are smaller than the marker size and are amplified by a factor 100 for visibility. The dashed red line is a cubic interpolation; the gold star marks the estimated maximum at $p^* \approx 1.33$ cm. The vertical green line indicates the Westinghouse 17×17 reference pitch of 1.26 cm.

Table 3: Infinite-medium multiplication factor k_∞ as a function of the lattice pitch for the 3×3 UO₂/H₂O assembly. Statistical uncertainties are quoted at 1σ and correspond to the combined collision/absorption/track-length estimator of MCNP6.

Pitch (cm)	k_∞	σ_{k_∞}
0.90	1.19250	0.00040
1.10	1.37507	0.00032
1.26	1.41856	0.00031
1.50	1.41092	0.00026
2.00	1.27576	0.00024

The shape of the curve is the textbook moderation behaviour qualitatively anticipated in Section 1. At $p = 0.90$ cm the moderator volume around each pin is so small that the resonance escape probability collapses: the multiplication factor drops to $k_\infty \simeq 1.19$, more than 0.2 below its maximum value. As the pitch is increased to $p = 1.10$ cm the spectrum starts to thermalise efficiently and k_∞ jumps to $\simeq 1.38$. Around $p = 1.26$ – 1.50 cm a broad plateau is visible, with k_∞ peaking at $\simeq 1.42$ and changing by less than 1% between the two extremes of this range. This flat optimum is a well-known feature of thermal reactor lattices and makes the multiplication factor relatively insensitive to small variations of the lattice pitch around its optimal value.

5.2 Estimation of the optimal pitch

To locate the maximum of the moderation curve more precisely than the discrete five-point scan allows, we interpolate the five MCNP data points with a cubic polynomial in p and identify the position and value of its peak. We call p^* the abscissa of the maximum (i.e. the estimated optimal lattice pitch) and $k_\infty^{\max}$ the corresponding value of the multiplication factor (i.e. the highest reactivity achievable with this lattice configuration). The cubic fit yields

$$p^* = 1.33 \text{ cm}, \quad k_\infty^{\max} = 1.430. \quad (5)$$

These two quantities are the ones we compare with the design value of a Westinghouse 17×17 assembly in the discussion below.

5.3 Comparison with the commercial PWR design

The agreement between the predicted optimum and the Westinghouse 17×17 design pitch of 1.26 cm is at the $\sim 5\%$ level. The dominant origin of this small offset is the room-temperature treatment of the moderator: a real PWR operates at $T \simeq 600$ K and 15 MPa, where the water density drops to $\rho_{\text{H}_2\text{O}} \simeq 0.71$ g/cm³, about 30% lower than the cold-water value used in this work. The reduced moderator density lowers k_∞ (less moderation per unit volume), an effect that we quantify explicitly in Section 6, where the reference run is repeated at the operating moderator density.⁴ A number of additional minor effects, not included in the present idealised model, would further contribute to the residual offset; their combined budget is consistent with the $\sim 5\%$ shift observed here, and therefore the result of Figure 2 can be considered in satisfactory agreement with the commercial PWR design.

⁴A full pitch scan at reduced density, which would be required to locate the shift of the moderation optimum, is beyond the scope of the present work.

6 Effect of moderator density (simplified temperature approach)

As anticipated in Section 5, the dominant simplifying assumption behind the $\sim 5\%$ offset between our calculated optimum $p^* \simeq 1.33$ cm and the Westinghouse design pitch $p = 1.26$ cm is the room-temperature treatment of the moderator. To quantify this effect we repeat the reference run of Section 4 with the moderator density reduced from $\rho_{\text{H}_2\text{O}} = 1.00$ g/cm³ to $\rho_{\text{H}_2\text{O}} = 0.71$ g/cm³, the typical operating value of a PWR primary circuit at $T \simeq 600$ K and 15 MPa. The mass fractions, the geometry and all the other physics options are left unchanged.

The multiplication factor obtained from the corresponding `kcode` run is

$$k_{\infty}^{\text{hot}}(p = 1.26 \text{ cm}, 3\%, \rho = 0.71) = 1.38847 \pm 0.00033, \quad (6)$$

to be compared with the cold-water value $k_{\infty}^{\text{cold}} = 1.41856 \pm 0.00031$ of equation (4). The relative reduction is

$$\frac{\Delta k_{\infty}}{k_{\infty}} = \frac{k_{\infty}^{\text{hot}} - k_{\infty}^{\text{cold}}}{k_{\infty}^{\text{cold}}} = -2.1\%, \quad (7)$$

that is, about half of the $\sim 5\%$ offset between our predicted optimum pitch and the commercial design value.

This single calculation therefore confirms that the moderator density is the dominant component of the temperature effect in our simplified model: it accounts on its own for roughly half of the discrepancy between the cold-lattice optimum and the design value of a real PWR. The fact that Δk_{∞} is *negative*, i.e. that a reduction of moderator density reduces the reactivity, is the physical origin of the self-stabilising behaviour of every PWR: any accidental rise of the moderator temperature lowers the water density, decreases k_{∞} , and brings the reactor back towards its equilibrium operating point. The remaining $\sim 2\text{--}3\%$ offset is largely expected to be covered by the actual spectral effects of the true operating temperature (which were neglected here by keeping the cross-sections at 300 K) and the other minor simplifications mentioned in Section 5.

7 Neutron Spectrum and Moderation Regimes

The moderation curve presented in Section 5 captures the macroscopic behaviour of the lattice but does not directly reveal the underlying neutron-population mechanism that produces it.

To make this physical mechanism visible we re-ran three of the `kcode` simulations of Section 5 – at $p = 0.90$, 1.26 and 2.00 cm – adding an `F4` flux tally on the central fuel pin. The tally records the average neutron flux in the cell as a function of energy, distributed over thirty-one logarithmically spaced bins from 10^{-9} to 20 MeV.

The three resulting spectra are plotted together in Figure 3, with the conventional partition of the energy axis in *thermal* ($E < 0.5$ eV), *epithermal* ($0.5 \text{ eV} < E < 0.1 \text{ MeV}$) and *fast* ($E > 0.1 \text{ MeV}$) ranges shaded in three different colours. The shape of the three spectra is a direct visualization of the moderation effect:

- at $p = 0.90$ cm the spectrum is markedly *hard*: the dominant contribution comes from the fast region around 1 MeV;

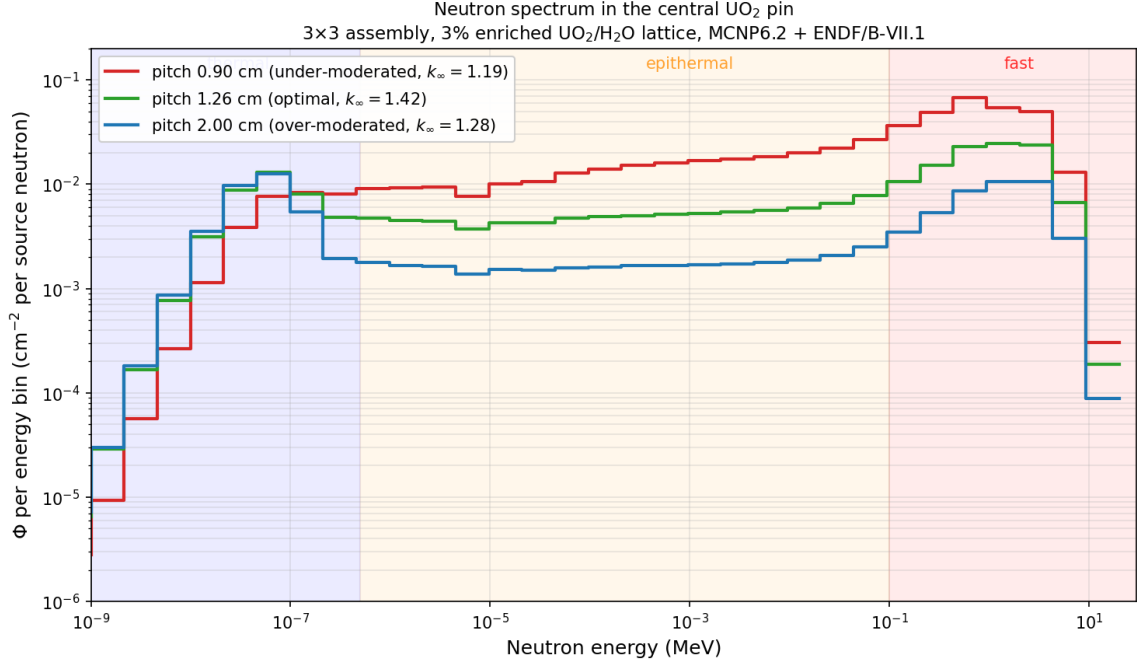


Figure 3: Neutron flux spectrum in the central UO_2 pin of the 3×3 assembly at three representative lattice pitches. The quantity plotted is the F4 tally integrated over each logarithmic energy bin, in units of cm^{-2} per source neutron per bin; the bin widths are equal in $\log E$ and span from 10^{-9} MeV to 20 MeV in 31 intervals. The thermal ($E < 0.5$ eV), epithermal and fast ($E > 0.1$ MeV) energy ranges are indicated by the shaded bands.

- at $p = 1.26$ cm the spectrum is *balanced*, with comparable thermal and fast contributions;
- at $p = 2.00$ cm the spectrum is strongly *thermalised*, with the fast component largely suppressed.

To turn this qualitative observation into a quantitative one we introduce the thermal-to-fast flux ratio

$$R = \frac{\Phi_{\text{thermal}}}{\Phi_{\text{fast}}} = \frac{\int_0^{0.5 \text{ eV}} \Phi(E) dE}{\int_{0.1 \text{ MeV}}^{20 \text{ MeV}} \Phi(E) dE}, \quad (8)$$

the values of which are reported in Table 4.

Table 4: Quantitative summary of the three spectra: integrated thermal flux ($E < 0.5$ eV), integrated fast flux ($E > 0.1$ MeV) and their ratio R . The ratio increases by almost a factor of eight between the under-moderated and the over-moderated case.

Pitch (cm)	Φ_{thermal}	Φ_{fast}	$R = \Phi_{\text{th}}/\Phi_{\text{fast}}$
0.90	2.94×10^{-2}	2.71×10^{-1}	0.108
1.26	3.87×10^{-2}	1.04×10^{-1}	0.372
2.00	3.42×10^{-2}	4.17×10^{-2}	0.821

These spectra provide a direct physical interpretation of the moderation curve of Section 5.

Under-moderated regime (p too small). The moderator around each fuel pin is so little that the neutrons born from fission do not have enough collisions with hydrogen to slow down to thermal energies. They remain in the epithermal range (~ 1 eV – 1 keV), exactly where ^{238}U has its strongest absorption resonances. A large fraction of them is therefore captured by ^{238}U before reaching thermal energies, the resonance escape probability p in the four-factor formula collapses, and k_∞ falls.

Over-moderated regime (p too large). The moderator now surrounds each pin in excess. Neutrons do reach thermal energies, but they spend a lot of their slowing-down history inside the water; before being captured by a ^{235}U nucleus, many of them are absorbed by hydrogen (which has its own non-negligible (n, γ) capture). The thermal utilisation factor f in the four-factor formula therefore drops, and k_∞ falls again – this time on the opposite side of the curve.

The intermediate configuration at $p = 1.26$ cm is where neither of the two effects dominates, and corresponds to the maximum of the moderation curve identified in Section 5.

8 Effect of the fuel enrichment

8.1 Setup of the enrichment scan

The third and last parametric study of this work concerns the dependence of the multiplication factor on the ^{235}U enrichment of the fuel. The lattice pitch is fixed at the reference value $p = 1.26$ cm, while the fraction of ^{235}U in the uranium component of the fuel is varied across four values which span the range of practical interest for thermal reactors:

- 1.5%: a low-end enrichment value, representative of fuels designed for low-power LWR systems;
- 3%: the reference Westinghouse 17×17 design, used as the baseline configuration in this work;
- 5%: the upper licensed limit for current commercial LWR fuel;
- 20%: the conventional boundary between low- and high-enriched uranium.

The mass fractions of the three uranium isotopes are recomputed for each enrichment from the molar masses of ^{235}U , ^{238}U and ^{16}O , while the oxygen content is left essentially unchanged at $\sim 11.87\%$ in mass. The other geometric, material and numerical settings are identical to those of the reference run of Section 4.

The four k_∞ values returned by the corresponding `kcode` runs are reported in Table 5.

Table 5: Infinite-medium multiplication factor k_∞ as a function of the ^{235}U enrichment for the 3×3 $\text{UO}_2/\text{H}_2\text{O}$ assembly at the reference pitch $p = 1.26$ cm. Statistical uncertainties are quoted at 1σ .

Enrichment (%)	k_∞	σ_{k_∞}
1.5	1.20611	0.00026
3.0	1.41856	0.00031
5.0	1.52263	0.00031
20.0	1.66622	0.00036

The dependence of k_∞ on the enrichment is monotonic but markedly non-linear. The largest gain in reactivity comes in the low-enrichment range: passing from 1.5% to 3% raises k_∞ by about 0.21, while doubling the enrichment again from 3% to 5% adds only 0.10, and the much larger jump up to 20% adds barely 0.14.

This saturation reflects the way the four factors of equation (1) respond to the depletion of ^{238}U from the fuel: as more ^{238}U is replaced by ^{235}U , the reproduction factor η_F improves quickly, the resonance escape probability p benefits from reduced ^{238}U absorption, and the thermal utilisation factor f approaches its limit. Above $\sim 5\%$ enrichment all these factors are close to their asymptotic values, and further increases of ^{235}U produce only marginal gains in k_∞ .

9 Master Summary

A compact view of all the results obtained in this work is presented in Table 6. The table collects the validation run on the original `Fision.txt` deck, the five runs of the pitch parametric study, the dedicated check at PWR operating temperature, and the four runs of the enrichment parametric study, together with the cubic-fit estimate of the optimum pitch.

10 Conclusions

The Monte Carlo simulations performed in this work illustrate, on a deliberately small but physically transparent system, the central role played by neutron moderation and fuel enrichment in determining the criticality of a thermal reactor lattice. Starting from the example deck `Fision.txt` provided with the course, which describes a fixed-source problem on two metallic-uranium rods in water, the geometry was extended to a 3×3 regular array of pins and the calculation mode was switched to a `kcode` criticality search; the use of reflective boundary conditions on the resulting unit cell turned the calculation into the standard lattice-physics setup that returns the infinite-medium multiplication factor k_∞ . The original `Fision.txt` deck was first validated by reproducing it on the host machine, with all the statistical convergence checks passed.

The 3×3 assembly at the Westinghouse reference pitch of 1.26 cm and 3% enrichment was found to have $k_\infty = 1.41856 \pm 0.00031$. The parametric study of the lattice pitch, performed at five values in the range 0.90–2.00 cm, returned the expected concave moderation curve, with a broad plateau between 1.26 and 1.50 cm and an estimated optimum at $p^* \simeq 1.33$ cm; the fall of k_∞ on either side of the optimum reproduces the qualitative

Table 6: Master summary of all the calculations. The first row gives the result of the validation run on the original `Fision.txt` deck (fixed-source, fission rate per source neutron). The next five rows give the infinite-medium multiplication factor k_∞ of the 3×3 $\text{UO}_2/\text{H}_2\text{O}$ assembly at five lattice pitches (3% enrichment); the line marked with $\star$ reports the cubic-fit estimate of the optimum pitch. The next four rows give k_∞ at the reference pitch $p = 1.26$ cm for four different enrichments. The last row reports the dedicated PWR-hot check at reduced moderator density.

Run	Configuration	p (cm)	Enr. (%)	Result
0	<code>Fision.txt</code> (fixed-source)	–	3.0	$R_f = (6.40 \pm 0.25) \times 10^{-3}$
1	3×3 , kcode (pitch scan)	0.90	3.0	$k_\infty = 1.19250 \pm 0.00040$
2	3×3 , kcode (pitch scan)	1.10	3.0	$k_\infty = 1.37507 \pm 0.00032$
3	3×3 , kcode (PWR ref.)	1.26	3.0	$k_\infty = 1.41856 \pm 0.00031$
4	3×3 , kcode (pitch scan)	1.50	3.0	$k_\infty = 1.41092 \pm 0.00026$
5	3×3 , kcode (pitch scan)	2.00	3.0	$k_\infty = 1.27576 \pm 0.00024$
$\star$	cubic-fit estimated optimum	1.33	3.0	$k_\infty^{\max} \simeq 1.430$
6	3×3 , kcode (enr. scan)	1.26	1.5	$k_\infty = 1.20611 \pm 0.00026$
7	3×3 , kcode (enr. scan, =3)	1.26	3.0	$k_\infty = 1.41856 \pm 0.00031$
8	3×3 , kcode (enr. scan)	1.26	5.0	$k_\infty = 1.52263 \pm 0.00031$
9	3×3 , kcode (enr. scan)	1.26	20.0	$k_\infty = 1.66622 \pm 0.00036$
10	3×3 , kcode (PWR-hot, $\rho_{\text{H}_2\text{O}} = 0.71$)	1.26	3.0	$k_\infty = 1.38847 \pm 0.00033$

behaviour predicted by the four-factor formula, with the resonance escape probability p dominating the loss budget at small pitch and the thermal utilisation factor f dominating it at large pitch. The estimated optimum is in good agreement with the commercial PWR design pitch of 1.26 cm. A dedicated check of the dominant simplifying assumption – the room-temperature treatment of the moderator – showed that reducing the water density to its typical PWR operating value of 0.71 g/cm³ accounts for roughly half of the 5% residual offset and yields a negative moderator-density coefficient, in line with the inherent safety features of a real LWR system.

The complementary parametric study of the fuel enrichment, performed at the reference pitch $p = 1.26$ cm in the range 1.5%–20%, returned a strongly non-linear, monotonically increasing dependence of k_∞ on the ²³⁵U content. The bulk of the reactivity gain is achieved in the low-enrichment range – passing from 1.5% to 3% enrichment k_∞ grows by 0.21, while the much larger step from 5% to 20% produces only an additional 0.14. The pronounced saturation observed at high enrichment reflects the fact that the four factors of the formula already approach their asymptotic values once the ²³⁸U content is significantly reduced.

A MCNP6 input decks used in this work

This appendix collects the MCNP6 input decks used to produce the results discussed in the main text. To keep the listing compact, only the reference deck (Section A.1) is reproduced in full; each subsequent parametric study is described in the form of the specific lines that need to be changed with respect to the reference deck (Sections A.2–

A.5). The full deck for any of the parametric variations can be reconstructed by taking the reference deck of Section A.1 and replacing the indicated lines with those given in the corresponding modification table.

A.1 Reference deck (`assembly_p126.txt`)

The full reference deck is available upon request. The structural modifications introduced with respect to the course example deck `Fision.txt` are: (i) extension of the geometry to a 3×3 pin array with nine `CZ` surfaces and a `RPP` bounding box; (ii) replacement of the metallic uranium material with UO_2 ceramic; (iii) addition of six reflective boundary conditions on all faces of the unit cell; (iv) replacement of the fixed-source card with a `kcode` criticality search. These changes are described in full in Section 2; the specific parameter values for each parametric run are given in the tables of Sections A.2–A.5.

A.2 Pitch parametric study (Section 5)

The five runs of the pitch scan use the same reference deck of Section A.1 with the lines listed in Table 7 replaced. Only the surfaces of the off-axis pin cylinders, the external box, and the initial fission source positions (`ksrc`) are modified.

Table 7: Lines of the reference deck modified for the five pitch values of Section 5. The cylinder surfaces 1–4 and 6–9 (off-axis pins) and the surface 200 (`RPP` box) and `ksrc` positions scale linearly with the pitch p . Surface 5 (central pin, `CZ 0.41`) and surfaces 100, 101 (`PZ` planes) are unchanged.

Pitch (cm)	Pin offset (cm)	RPP half-side (cm)	ksrc x,y values (cm)
0.90	± 0.90	± 1.35	$\pm 0.90, 0.00, \pm 0.90$
1.10	± 1.10	± 1.65	$\pm 1.10, 0.00, \pm 1.10$
1.26	± 1.26	± 1.89	$\pm 1.26, 0.00, \pm 1.26$
1.50	± 1.50	± 2.25	$\pm 1.50, 0.00, \pm 1.50$
2.00	± 2.00	± 3.00	$\pm 2.00, 0.00, \pm 2.00$

A.3 Enrichment parametric study (Section 8)

The four runs of the enrichment scan use the reference deck of Section A.1 (with $p = 1.26$ cm) and modify only the fuel material card `m100`. The mass fractions of the three nuclides for each enrichment are listed in Table 8.

Table 8: Mass fractions of the three nuclides composing UO_2 for the four enrichment values studied in Section 8. Only the `m100` card is changed; everything else in the deck of Section A.1 stays unchanged.

Enrichment (%)	$m_{235\text{U}}$	$m_{238\text{U}}$	$m_{16\text{O}}$
1.5	0.01305	0.86847	0.11848
3.0	0.02645	0.85490	0.11865
5.0	0.04354	0.83789	0.11857
20.0	0.17446	0.70678	0.11876

A.4 Spectrum tally addition (Section 7)

The three runs of the spectrum study (at pitches 0.90, 1.26 and 2.00 cm) use the reference deck of Section A.1, with the geometry modifications of Table 7 applied where appropriate, and add the following **F4** flux tally with logarithmic energy binning at the end of the data cards block:

```
c --- TALLIES ---
f14:n 5
sd14 5.281
e14 1e-9 30ilog 20
```

The **sd14** card explicitly supplies the volume of the central fuel pin ($V = \pi r_p^2 h \simeq 5.281 \text{ cm}^3$) because MCNP cannot compute it automatically when reflective boundary conditions are imposed on the bounding planes.

A.5 Moderator temperature variation (Section 6)

The single run that simulates PWR operating temperature (Section 6) uses the reference deck of Section A.1 with the cell card for the moderator modified to lower its density from 1.00 to 0.71 g/cm³. Only the line defining cell 10 is changed:

```
10 200 -0.71 +1 +2 +3 +4 +5 +6 +7 +8 +9
      -200 -100 +101 imp:n=1
```

The composition of material 200 (light water) and the thermal scattering library are kept unchanged: only the macroscopic density is reduced.

References

- [1] J. R. Lamarsh, *Introduction to Nuclear Engineering*, 2nd ed., Addison-Wesley, 1984.
- [2] S. Glasstone, A. Sesonske, *Nuclear Reactor Engineering*, 4th ed., Van Nostrand Reinhold, 1994.
- [3] C. J. Werner et al., *MCNP version 6.2 Release Notes*, Los Alamos National Laboratory, LA-UR-18-20808 (2018).
- [4] M. B. Chadwick et al., “ENDF/B-VII.1 Nuclear Data for Science and Technology: Cross Sections, Covariances, Fission Product Yields and Decay Data,” *Nuclear Data Sheets* **112**, 2887–2996 (2011).
- [5] J. J. Duderstadt, L. J. Hamilton, *Nuclear Reactor Analysis*, John Wiley & Sons, 1976.
- [6] I. Porras, J. Praena, A. Ibarra, *Tecnología Nuclear – Lecture notes*, Master in Physics, Universidad de Granada, 2025/2026.